$$
\begin{aligned}
C_{\phi G}^{(11)} \to \frac{1}{16\,\pi^2} \\
& \Big(-\frac{1}{24}\sum_p g_1{}^2\, g_3{}^2\, LF_{3,0}\big[m_{\tilde d}{}^p\big] + \frac{1}{24}\sum_p g_1{}^2\, g_3{}^2\, LF_{4,-1}\big[m_{\tilde d}{}^p\big] - \frac{1}{24}\, g_3{}^2 \left(6\,\overline{y_u}^{pr}\, y_u{}^{pr} + \sum_p g_1{}^2\right) LF_{3,0}\big[m_{\tilde q}{}^p\big] + \\
& \quad \frac{1}{24}\, g_3{}^2 \left(6\,\overline{y_u}^{pr}\, y_u{}^{pr} + \sum_p g_1{}^2\right) LF_{4,-1}\big[m_{\tilde q}{}^p\big] + \frac{1}{12}\sum_p g_1{}^2\, g_3{}^2\, LF_{3,0}\big[m_{\tilde u}{}^p\big] - \\
& \quad \frac{1}{12}\sum_p g_1{}^2\, g_3{}^2\, LF_{4,-1}\big[m_{\tilde u}{}^p\big] - \frac{1}{4}\, g_3{}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{3,0}\big[m_{\tilde u}{}^r\big] + \frac{1}{4}\, g_3{}^2\, \overline{y_u}^{pr}\, y_u{}^{pr}\, LF_{4,-1}\big[m_{\tilde u}{}^r\big] - \\
& \quad \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{2,2,0}\big[m_{\tilde d}{}^r,\, m_{\tilde q}{}^p\big] - \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{3,1,0}\big[m_{\tilde d}{}^r,\, m_{\tilde q}{}^p\big] + \\
& \quad \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{3,2,-1}\big[m_{\tilde d}{}^r,\, m_{\tilde q}{}^p\big] + \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{4,1,-1}\big[m_{\tilde d}{}^r,\, m_{\tilde q}{}^p\big] - \\
& \quad \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{3,1,0}\big[m_{\tilde q}{}^p,\, m_{\tilde d}{}^r\big] + \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{3,2,-1}\big[m_{\tilde q}{}^p,\, m_{\tilde d}{}^r\big] + \\
& \quad \frac{1}{4}\, g_3{}^2\, \tilde{\mu}^2\, \overline{y_d}^{pr}\, y_d{}^{pr}\, LF_{4,1,-1}\big[m_{\tilde q}{}^p,\, m_{\tilde d}{}^r\big] - \frac{1}{4}\, g_3{}^2\, \overline{a_u}^{pr}\, a_u{}^{pr}\, LF_{2,2,0}\big[m_{\tilde q}{}^p,\, m_{\tilde u}{}^r\big] - \\
& \quad \frac{1}{4}\, g_3{}^2\, \overline{a_u}^{pr}\, a_u{}^{pr}\, LF_{3,1,0}\big[m_{\tilde q}{}^p,\, m_{\tilde u}{}^r\big] + \frac{1}{4}\, g_3{}^2\, \overline{a_u}^{pr}\, a_u{}^{pr}\, LF_{3,2,-1}\big[m_{\tilde q}{}^p,\, m_{\tilde u}{}^r\big] + \\
& \quad \frac{1}{4}\, g_3{}^2\, \overline{a_u}^{pr}\, a_u{}^{pr}\, LF_{4,1,-1}\big[m_{\tilde q}{}^p,\, m_{\tilde u}{}^r\big] - \frac{1}{4}\, g_3{}^2\, \overline{a_u}^{pr}\, a_u{}^{pr}\, LF_{3,1,0}\big[m_{\tilde u}{}^r,\, m_{\tilde q}{}^p\big] + \\
& \quad \frac{1}{4}\, g_3{}^2\, \overline{a_u}^{pr}\, a_u{}^{pr}\, LF_{3,2,-1}\big[m_{\tilde u}{}^r,\, m_{\tilde q}{}^p\big] + \frac{1}{4}\, g_3{}^2\, \overline{a_u}^{pr}\, a_u{}^{pr}\, LF_{4,1,-1}\big[m_{\tilde u}{}^r,\, m_{\tilde q}{}^p\big]\Big)
\end{aligned}
$$